

A calibrated, auditable deployment layer for forensic surface comparison

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The thesis. Verity is not a new matching algorithm. It is the **production, calibration, and validation layer** for the CSAFE / Iowa State research program (Hare, Hofmann & Carriquiry 2017) and the NIST congruent-matching line (Song 2013). It reports a **calibrated weight of evidence** (a likelihood ratio) with a **characterized cost** (C_{llr}) on a **named** reference population — and refuses to emit an answer outside the domain it was validated on. It does *not* claim a universal error rate, because none exists.

The problem

The NRC (2009) and PCAST (2016) reports found that feature-comparison forensic disciplines — firearms and toolmarks among them — lack *characterized error rates*. Cuellar et al. (2024) went further: *no* black-box study of firearm comparison has properly characterized one, largely because studies test on the same sources they were tuned on and report study-specific numbers as if they were properties of the whole discipline. The field needs a defensible, reproducible way to state how strong firearm/toolmark evidence is — one that survives scrutiny.

Verity's answer

- **Report a likelihood ratio, not a verdict.** How much more probable the observed surface agreement is under same-source than different-source — prior-independent, the form courts and statisticians ask for.
- **Characterize it honestly.** Quote the calibration cost C_{llr} (Brümmer & du Preez 2006) measured *source-disjoint* (no firearm in both training and test), reported *per study* — never pooled across makes for error-rate claims.
- **Know your limits.** An applicability-domain gate declines to score a scan outside the validated domain instead of extrapolating.

What is genuinely new (and what is borrowed)

Built on (credited): the CSAFE bullet-matching method and `bulletxtrctr`; Song's Congruent Matching Cells for cartridge cases; the Chumbley toolmark statistic; the ISO 25178-72 X3P format and `x3ptools`; the C_{llr} metric from forensic speaker recognition.

New in Verity:

1. **Unification** via *Congruent Matching Regions* — one algorithm across striated, impressed, and fractured marks.
2. A **calibrated-LR answer** to Cuellar et al.
3. A **glass-box** decision layer: region attribution + an *audit firewall*.
4. **Production engineering:** native Rust X3P codec, deployed API.
5. A **data-selected scorer** (`diag_contrast`).

How it works

One pipeline, any mark. ISO-standard surface metrology (degree-2 form removal; ISO 16610 roughness band, $\lambda_s = 4\mu\text{m}$, $\lambda_c = 250\mu\text{m}$) → FFT-based striae orientation and signature extraction → comparison → a calibrated likelihood ratio. **Congruent Matching Regions** generalizes Song's cell-counting: partition one surface into regions, register each against the other, count the regions whose best-fit transforms agree. A 1-D window under lag parallels the Chumbley striae statistic; a 2-D cell under translation+rotation instantiates CMC; a 3-D patch handles fractures. **The audit firewall** turns any score into an LR through a *monotone, bounded* calibration (Platt scaling, capped at what the reference supports, with a bootstrap interval) — so the evidence is interpretable no matter how the score was produced, and the system reports *which regions* drove the match.

The evidence

On the Hamby-252 consecutively-rifled benchmark, under the source-disjoint protocol that directly answers Cuellar et al.:

Barrel-disjoint test $C_{llr} = 0.113 \pm 0.066$ | **test AUC** $= 1.000 \pm 0.001$

(full-study totals: 46 same-source / 549 different-source pairs across 10 barrels; each fold tests only the pairs among its four held-out barrels; in-sample $C_{llr} = 0.077$, $C_{llr}^{\min} = 0.032$, ECE = 0.031, EER = 2.1 %.)

Table 1. Source-disjoint test C_{lr} per study (lower is better); **bold** = better scorer. Reported per study, never pooled.

Study	diag_mean	diag_contrast
Hamby-252	0.121	0.113
PGPD Beretta	0.299	0.273
Phoenix Ruger	0.252	0.354
Hamby-173	0.456	0.338

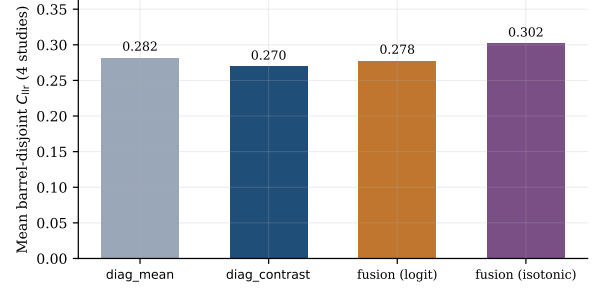


Figure 1. Scorer chosen *by data*: **diag_contrast** has the lowest cross-study source-disjoint C_{lr} ; adding fusion capacity does not help, and the isotonic fusion overfits.

The platform

- Native **Rust X3P codec** (ISO 25178-72) with Python & R bindings — fast, dependency-light, interoperable with **x3ptools**.
- A **deployed calibrated-LR REST API**: `/detect`, `/compare` (bounded LR + credible interval + region attribution), `/health` (publishes the calibrated domains).
- A **data catalog** that harvests/labels the open NBTRD 3-D scans, and a **reproducible validation harness** behind every number here.

Honest scope

- The scorer and front-end were selected on the same four studies reported here; a one-shot confirmation on untouched data is the next milestone.
- Consecutively-manufactured benchmarks are small in *sources* (8–10 barrels; Fadul ≈ 10 slides), so fold-to-fold C_{lr} varies.
- A score-based LR is a property of the score and the named reference, not an unconditional measure — hence the scope gate.
- **No universal error rate** is claimed; every number is conditioned on a named population and a stated protocol.
- The learned (Phase-2b) representation does not yet beat the interpretable baseline; the baseline is deployed.

An invitation, not a claim of endorsement. Verity is the deployment and calibration layer for a research program it did not invent. The next step is validation on larger, representative, genuinely source-disjoint references — the data the CSAFE program is built to produce. The codec, the calibration firewall, and the validation harness are offered as open infrastructure for that collaboration. Mention of CSAFE, Iowa State, or NIST denotes intellectual lineage and a prospective partnership only; it does not imply review or endorsement.